

Assignment - 04

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Sec : 19

CSE340

Ans. to the ques. no. 1

Add X_{21}, X_{22}, X_{23}

For single cycle -

$$\begin{aligned} & F + \text{Reg Read} + \text{ALU} + \text{Reg Write} \\ &= 20 + 20 + 40 + 20 \\ &= 100s \end{aligned}$$

For pipeline -

$$\begin{aligned} \text{Highest execution time among stages} &= 40s \\ &= 40 \times 5 = 200s \end{aligned}$$

That is,

^{Single cycle}
~~Pipelined~~ setup is faster than ^{pipelined} ~~single cycle~~ setup.

Ans. to the ques. no. 2

i) ALU Src = 0

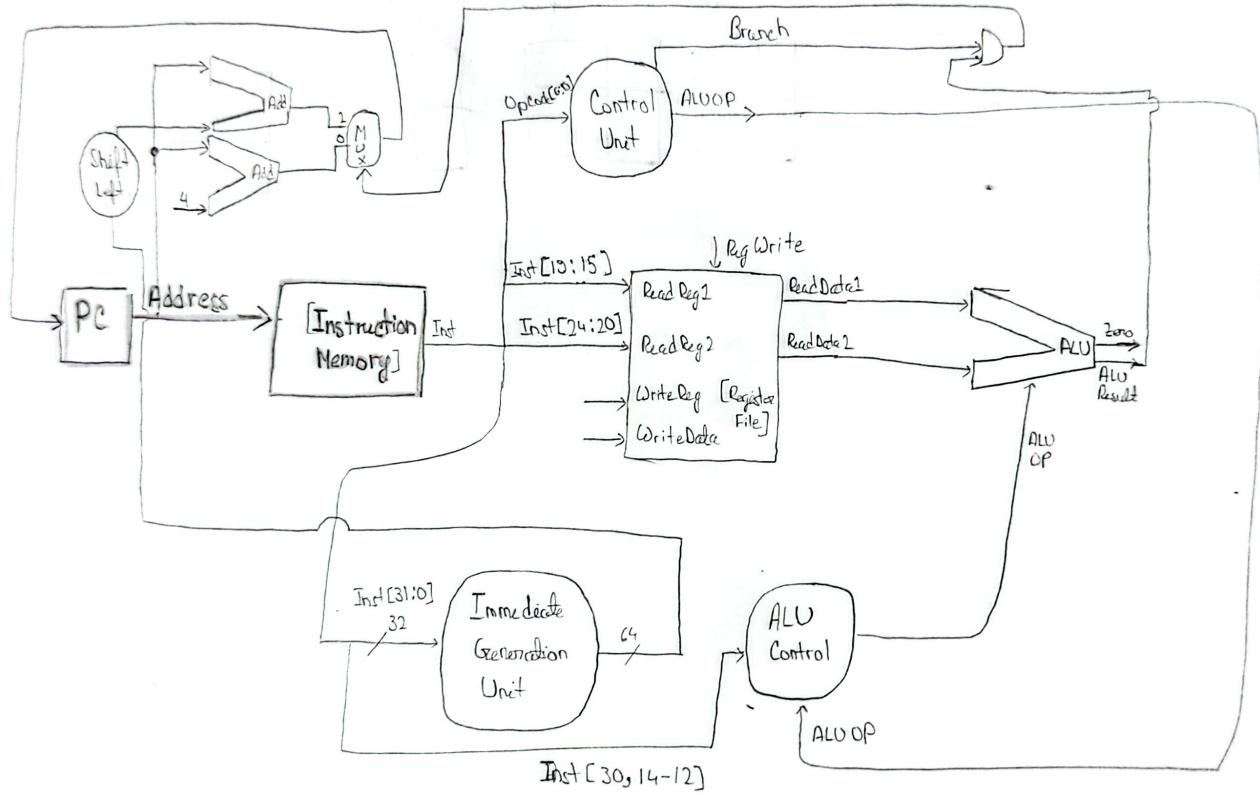
ii) RegWrite = 1

iii) Mem Read = 1

iv) Mem Write = 0

v) MemtoReg = 00

FIGURE NO.



Ans. to the ques. no. 3

Ans. to the ques. no. 4

In pipelined setup, we need to access memory two times: one for fetching the instruction and another for read/write data. If there was a combined memory for both these stages, the F and M stage would conflict when trying to access it simultaneously and it would cause structural hazard. For we know that, one instruction uses only one resource in each clock cycle. This would happen on every Load/store instruction. By keeping the memory separated, F and M stage can proceed parallelly ~~and no~~ without having any conflict and there would be no structural hazard from memory access as well as no forced stalling.

Ans. to the ques. no. 5

Yes there will be data hazard during execution.

CC1 CC2 CC3 CC4 CC5 CC6 CC7 CC8 CC9 CC10 CC11

i) F D E M WB

ii) F D E M WB

iii) F D → E M WB

X X X X X

iv) F D → E M WB

v) F D → E M WB

vi) F D → E M WB

$$\text{Average CPI} = \frac{\text{CC needed}}{\text{IC}}$$

$$= \frac{11}{6} = 1.833.$$